

# International Certifications Portfolio

## GADZIDE Komi Joël – CSC Scholarship Application 2026

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## EXECUTIVE SUMMARY

This portfolio presents [9 international certifications](#) from leading universities, representing 300+ hours of self-directed study validated through 20+ technical projects. The combination of theoretical knowledge from Duke University, University of Michigan, University of Illinois, and UC Irvine with practical implementation across multiple programming domains demonstrates exceptional readiness for advanced software engineering education and meaningful contributions to China-Africa technological cooperation.

**Key Achievement:** All certifications completed independently while maintaining Baccalauréat(High Secondary School Diploma) ,Mention Très Bien(With Honor), Grade :16/20 ; demonstrating exceptional self-motivation and technical aptitude.

**Strategic Application:** These skills directly support my goal to develop autonomous systems (Multi-purpose drones and Robots, smart home solutions) for for Africa's technological advancement through Sino-African cooperation.

## Certifications Summary

| # | Certificate                                         | Institution                       | Completed | Hours | Key Skills                                                                | GitHub Projects                                              |
|---|-----------------------------------------------------|-----------------------------------|-----------|-------|---------------------------------------------------------------------------|--------------------------------------------------------------|
| 1 | Programming Foundations with JavaScript, HTML & CSS | Duke University (Coursera)        | 2025      | 32    | JavaScript, HTML/CSS, DOM manipulation, web development                   | <a href="#">Plant Store Landing Page</a>                     |
| 2 | Programming Fundamentals                            | Duke University (Coursera)        | 2025      | 18    | Core programming logic, computational thinking, problem-solving           | <a href="#">Tic-Tac-Toe GUI Game</a> (Python Tkinter)        |
| 3 | Programming for Everybody (Python)                  | University of Michigan (Coursera) | 2025      | 18    | Python fundamentals, data types, loops, file I/O                          | <a href="#">Numerical Analysis Notebooks</a> (Multiple)      |
| 4 | Object-Oriented Data Structures in C++              | University of Illinois (Coursera) | 2025      | 21    | C++ OOP, data structures (linked lists, trees, stacks), memory management | <a href="#">C Linked List Operations, C++ Banking System</a> |

|   |                                                         |                                            |      |     |                                                                                                            |                                                                                                                   |
|---|---------------------------------------------------------|--------------------------------------------|------|-----|------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| 5 | <b>Introduction to IoT &amp; Embedded Systems</b>       | University of California Irvine (Coursera) | 2025 | 11  | IoT architecture, embedded hardware, system design, protocols                                              | <a href="#">Arduino Serial LED Projects</a> (2 projects)                                                          |
| 6 | <b>Java Programming: Solving Problems with Software</b> | Duke University (Coursera)                 | 2025 | 18  | Java fundamentals, OOP principles, algorithms, problem-solving                                             | <a href="#">Library Management System</a> , <a href="#">Pokemon Battle System</a> , <a href="#">Store Project</a> |
| 7 | <b>Mastering Python Sequences and Strings</b>           | EDUCBA (Coursera)                          | 2025 | 7   | Python data structures optimization, string manipulation, performance                                      | <a href="#">Applied Numerical Methods</a> , <a href="#">Matrix Analysis</a>                                       |
| 8 | <b>Machine Learning with Python</b>                     | FreeCodeCamp                               | 2025 | 100 | Supervised/unsupervised learning, scikit-learn, neural networks, model evaluation                          | <a href="#">ML Implementations</a> , <a href="#">Data Analysis Projects</a>                                       |
| 9 | <b>The Arduino Platform and C Programming</b>           | University of California Irvine (Coursera) | 2025 | 12  | C Programming, Embedded Software, Microcontrollers (Arduino), Hardware Interfacing, Computational Thinking | <a href="#">C Linked List Operations</a>                                                                          |

**Total:** 220+ documented hours (additional 80+ hours in project development and debugging)

**Note:** All certificates are verified with unique platform IDs and available for inspection upon request.

## Technical Projects Breakdown

### Programming Fundamentals & Web (3 projects)

- **Plant Store Landing Page** (HTML/CSS/JavaScript) – Responsive web design
- **Tic-Tac-Toe GUI** (Python Tkinter) – Event-driven programming
- **Multiple web components** – DOM manipulation, interactivity

### Object-Oriented Programming (6+ projects)

**Java:**

- **Library Management System** – Complete CRUD with inheritance
- **Pokemon Battle System** – Abstract classes, polymorphism
- **Movie Recommendation System** – Data analysis algorithms
- **Store Management** – Inventory and transaction handling

**C++:**

- **Banking Management System** – Advanced OOP with file I/O
- **Scientific Calculator** – Expression parsing, data structures
- **Mini Compiler** – Lexer, parser, interpreter implementation

- **Snake Game** – Real-time game loop, collision detection

## **Data Structures & Algorithms and The Arduino Platform & C Programming (4+ projects)**

- **C Linked List Operations** – Manual memory management
- **Network Optimization (Dijkstra's Algorithm)** – Graph theory
- **Graph Representation & Adjacency** – Multiple implementations
- **Custom data structures** – Used across all C++ projects

## **Python Scientific Computing (10+ projects)**

- **Numerical Analysis with SymPy** – Symbolic mathematics
- **Applied Kinematics & Integration** – Physics simulations
- **Particle Kinematics & Arc Length** – Computational physics
- **Matrix Analysis & Linear Systems** – Linear algebra
- **Multivariable Function Visualization** – 3D graphing
- **Applied Numerical Methods** – Calculus and algebra
- **Multiple Jupyter notebooks** – Scientific computing workflows

## **Embedded Systems & IoT (2 projects)**

- **Arduino Serial LED Control (Project 1)** – Serial communication
- **Arduino Serial LED Control with Web Interface (Project 2)** – IoT integration

## **Game Development (2 projects)**

- **Endless Car Runner** (Python Pygame) – Game mechanics, sprites
- **Snake Game** (C++ Console) – Real-time input, collision detection

**GitHub Stats:** 20+ repositories | 6 programming languages | 100+ commits

## **Certification Details: Course Requirements & Practical Validation**

Each certification required specific technical deliverables and was validated through rigorous projects demonstrating mastery beyond theoretical knowledge.

### **Domain 1: Software Engineering & Object-Oriented Programming**

#### ***1- Java Programming: Solving Problems with Software (Duke University)***

- **Requirement:** Design and implement complete systems using OOP principles (inheritance, polymorphism, encapsulation)
- **Score:** 88.31% (Passed with Honors)
- **Validation:** Mastery of complex Data Structures (HashSet, HashMap) and encryption algorithms.
- **Extended Applications(Post-Completion) :** Developed three production-quality Java systems:
- **Library Management System** – Demonstrates class hierarchies and data persistence

- **Pokemon Battle System** – Abstract base classes with type-based polymorphism
- **Store Management** – Real-world business logic with CRUD operations
- **GitHub:** All source code fully documented and available

## ***2- Object-Oriented Data Structures in C++ (University of Illinois)***

- **Requirement:** Implement complex data structures (linked lists, trees, hash tables) with manual memory management
- **Score:** 96%
- **Validation:** Data structures (lists, trees) using C++ Object-Oriented Programming (OOP)
- **Extended Applications(Post-Completion) :**
- **C Linked List Operations** – Manual pointer manipulation, memory allocation
- **C++ Banking System** – STL containers, smart pointers, file I/O
- **C++ Projects** – Applied DS concepts across multiple projects
- **Skills Demonstrated:** Low-level memory management, STL mastery, performance optimization

## **Domain 2: Programming Fundamentals & Logic**

### ***3- Programming Fundamentals (Duke University)***

- **Score:** 97.50% (Honors)
- **Requirement:** Successfully complete computational thinking challenges and algorithmic mini-projects
- **Validation:** Computational Thinking challenges and mini-projects requiring the application of algorithms.
- **Extended Applications(Post-Completion) :** Tic-Tac-Toe GUI game demonstrating structured problem-solving approach
- **Impact:** Foundation for all subsequent algorithmic work

### ***4- Programming for Everybody – Python (University of Michigan)***

- **Score:** 98.25% (Top Performance)
- **Requirement:** File processing, data parsing, and analysis exercises
- **Validation :** File processing and basic data analysis (Data Parsing)
- **Extended Applications(Post-Completion) :** Multiple Python projects including:
- Numerical analysis notebooks
- Data manipulation workflows
- Scientific computing applications
- **Impact:** Python became primary language for scientific computing

### ***5- Mastering Python Sequences and Strings (EDUCBA)***

- **Requirement:** Deep understanding of Python data structures for performance optimization
- **Score:** 87.50%
- **Validation:**
- Optimized data handling across all Python projects
- Python sequential objects (lists, tuples, dictionaries, sets) to ensure high performance and efficiency of data structures

- **Application:** Used in every Python notebook for efficient data preparation and algorithmic implementations

#### ***6- The Arduino Platform and C Programming (UC Irvine)***

- **Score:** 97.50% (Honors)
- **Requirement:** Successfully complete computational thinking challenges and algorithmic mini-projects
- **Validation: Computational Thinking challenges and mini-projects requiring the application of algorithms.**
- **Extended Applications(Post-Completion) :** Tic-Tac-Toe GUI game demonstrating structured problem-solving approach.
- **Impact:** Foundation for all subsequent algorithmic work

### **Domain 3: Artificial Intelligence & Machine Learning**

#### ***7- Machine Learning with Python (FreeCodeCamp)***

- **Requirement:** Complete 5 mandatory ML projects including classification, regression, and neural networks
- **Hours:** 100 (most intensive certification)
- **Validation:**
- Implemented supervised/unsupervised learning algorithms
- Built, trained, and validated multiple ML models
- Applied to real datasets with performance evaluation
- **Relevance:** Foundational for autonomous systems (drones, smart homes) requiring AI capabilities

### **Domain 4: Web Development & Computational Thinking,**

#### ***8- Programming Foundations with JS, HTML and CSS (DUKE University)***

- **Requirement:** Design and implement foundational web applications, focusing on event-driven programming, image processing, and algorithms for interactive user interfaces.
- **Score:** 86.42%
- **Validation:**
- Computational Thinking: Proven capacity to design programming solutions using core concepts (functions, loops, conditionals).
- Algorithms & Image Analysis: Successful implementation of algorithms for dynamic image manipulation, including creating filters (e.g., grayscale, green screen effect) via JavaScript pixel processing.
- Interactive Design: Mastery of Event-Driven Programming, connecting user actions to dynamic responses on the webpage.
- Web Structure (UI/UX): Competence in building structured web pages using proper HTML elements (paragraphs, links, images) and applying effective CSS styling (using IDs and classes).
- **Skills:** JavaScript, HTML/CSS, DOM manipulation, User Interface (UI) Design, web development

- **Extended Applications(Post-Completion)** : Plant Store Landing Page (A project demonstrating competence in responsive front-end design, directly utilizing the HTML/CSS/JavaScript skills acquired.)

## Domain 5: Embedded Systems & Internet of Things

### 9- Introduction to IoT & Embedded Systems (UC Irvine)

- **Requirement:** Design complete embedded system architecture and analyze hardware-software trade-offs
- **Score:** 91%
- **Validation:**
  - Theoretical design of IoT systems
  - Understanding of protocols (MQTT, HTTP, I2C)
  - Real-time sensor integration
  - Applied in Arduino projects
- **Skills:** Debugging embedded systems, interrupt handling, hardware constraints
- **Real-World Extension:** Enhanced by intensive 1-month phone repair experience (hardware troubleshooting, component-level understanding)

## Skills Matrix

| Skill Category               | Proficiency Level | Certifications | Projects |
|------------------------------|-------------------|----------------|----------|
| Object-Oriented Programming  | Advanced          | 1              | 6+       |
| Python Scientific Computing  | Advanced          | 2              | 10+      |
| Data Structures & Algorithms | Intermediate      | 3              | 5+       |
| Embedded Systems & IoT       | Intermediate      | 1              | 2+       |
| Machine Learning             | Intermediate      | 1              | Multiple |
| Web Development              | Intermediate      | 1              | 2+       |
| Game Development             | Intermediate      | -              | 3        |

## Impact & Relevance for CSC Scholarship

### Academic Preparedness

These certifications provide comprehensive preparation for university-level computer science and software engineering programs:

- **Theoretical Foundation:** Strong understanding of CS fundamentals, algorithms, and system design
- **Practical Skills:** 20+ completed projects demonstrating ability to build production-quality software
- **Self-Learning Ability:** 300+ hours of independent study proves motivation and discipline for rigorous academic programs

## Research Potential

Skills directly applicable to cutting-edge research areas:

- **Robotics:** Embedded systems + algorithms + AI/ML
- **IoT Solutions:** Network programming + hardware integration
- **Smart Systems:** ML models + embedded deployment + real-time processing

## Development Impact for Togo

Technical foundation to address local challenges:

- **Agricultural Technology:** ML for crop monitoring + IoT sensors + drone control systems
- **Smart Infrastructure:** Embedded systems for smart homes adapted to African power/connectivity constraints
- **Education:** Share knowledge through workshops, mentor young programmers

## China-Africa Technology Bridge Potential

This technical foundation positions me to facilitate meaningful technological cooperation:

- **Adaptation Capability:** Experience in resource-constrained computing prepares me to adapt Chinese technologies to African contexts
- **Knowledge Transfer:** Comprehensive understanding of both theoretical CS and practical engineering enables effective technology transfer
- **Innovation Bridge:** Ability to identify Chinese technologies with high potential for African adaptation and deployment
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## Verification & Authenticity

- ✓ All certifications include unique verification IDs
- ✓ Digital certificates available upon request (on : <https://github.com/gadzide-joel/Portfolio-projects/tree/main/CERTIFICATIONS>)
- ✓ Coursera profiles publicly viewable
- ✓ GitHub repository is public: <https://github.com/gadzide-joel/Portfolio-projects>
- ✓ Course completion emails and grade reports available

**Attestation:** I, GADZIDE Komi Joël, certify that all information provided is accurate and verifiable. I completed each certification independently through legitimate online platforms.

## Contact Information

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**Application:** Chinese Government Scholarship (CSC) 2026 – Bachelor's Degree in Software Engineering

## **STRATEGIC VALUE FOR SINO-AFRICAN COOPERATION**

This portfolio demonstrates not just course completion, but practical mastery through 20+ implemented projects spanning web development, embedded systems, machine learning, and scientific computing. The combination of theoretical knowledge from prestigious international institutions and hands-on project experience demonstrates:

1. Exceptional Readiness for advanced software engineering studies in China
2. Proven Capacity to bridge theoretical knowledge with practical implementation
3. Direct Relevance to China's leadership in AI, IoT, and large-scale systems
4. Tangible Potential to contribute to technological cooperation between China and African nations

## **CONCLUSION**

This portfolio provides concrete evidence of technical capability, self-learning discipline, and research potential - essential qualities for success in China's rigorous academic environment and for contributing to Sino-African technological cooperation. The quantifiable achievements (300+ hours, 8 certifications, 20+ projects) demonstrate not just ambition but proven ability to execute complex technical work independently.

The combination of theoretical knowledge from world-class institutions and practical implementation across multiple programming domains creates a unique profile perfectly aligned with China's focus on applied technological innovation and international educational cooperation.